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Date : May 2026

SQL JOINS (EXERCISES)

@BrightLearn

PART A : INNER JOIN

Below i have attatched the Screenshots of the queries and outputs of 5 questions from part A :

The screenshot shows the Databricks SQL Editor interface. The SQL query is as follows:

```
74 INSERT INTO viewing_sessions VALUES (905, 3, 705, 90);
75
76
77 --PART A : INNER JOIN--
78 --Question 1: SHOW EVERY USER WHO A SUBSCRIPTION
79 SELECT users.user_id,
80        users.user_name,
81        subscriptions.subscription_id,
82        subscriptions.start_date
83 FROM users
84 INNER JOIN subscriptions
85 ON users.user_id = subscriptions.user_id;
86
```

The output table shows 4 rows:

user_id	user_name	subscription_id	start_date
1	Nomvula	501	2026-01-15
2	Nomvula	503	2026-03-10
3	David	502	2026-02-01
4	Anele	505	2026-04-05

4 rows | 0.31s runtime | See performance (1) | Optimize

Question 2:

The screenshot shows the Databricks SQL Editor interface. The SQL query is as follows:

```
92 SELECT *
93 FROM subscriptions;
94
95 --Question 2: Show every subscription with its matching plan_name and monthly price
96 --I only bypass by reference the columns which appear to be ambiguous
97 SELECT subscriptions.plan_id,
98        subscriptions.plan_name,
99        subscriptions.monthly_price
100 FROM subscriptions
101 INNER JOIN plans
102 ON subscriptions.plan_id=plans.plan_id;
103
104
```

The output table shows 5 rows:

plan_id	subscription_id	plan_name	monthly_price
11	502	Standard	129.00
11	504	Standard	129.00
12	503	Premium	199.00
13	505	Family	249.00
10	501	Basic	75.00

5 rows | 1.53s runtime | See performance (1) | Optimize

Question 3 :

The screenshot shows the Databricks SQL Editor interface. The query editor contains the following SQL code:

```
105 SELECT *
106 FROM viewing_sessions;
107
108 -- Question 3 : Show every Viewing session that has a matching show
109 SELECT session_id,
110        user_id,
111        show_title,
112        genre,
113        watch_minutes
114 FROM viewing_sessions
115 INNER JOIN shows
116 ON viewing_sessions.show_id=shows.show_id;
```

The results table displays 4 rows of data:

session_id	user_id	show_title	genre	watch_minutes
902	2	Tech Tales	Documentary	30
904	7	Comedy Hour	Comedy	20
901	1	Comedy Hour	Comedy	45
903	1	Crime Time	Drama	60

At the bottom, it indicates "4 rows | 2.43s runtime" and "See performance (1)".

Question 4 :

The screenshot shows the Databricks SQL Editor interface. The query editor contains the following SQL code:

```
115 INNER JOIN shows
116 ON viewing_sessions.show_id=shows.show_id ;
117
118 -- Question 4 : Show every viewing session with user who watched it
119 SELECT user_name,
120        country,
121        session_id,
122        show_id,
123        watch_minutes
124 FROM users
125 INNER JOIN viewing_sessions
126 ON users.user_id=viewing_sessions.user_id;
```

The results table displays 4 rows of data:

user_name	country	session_id	show_id	watch_minutes
Nomvula	Johannesburg	903	702	60
Nomvula	Johannesburg	901	701	45
David	Cape Town	902	703	30
Anele	Durban	905	705	90

At the bottom, it indicates "4 rows | 1.85s runtime" and "See performance (1)".

Question 5 :

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Data Engineering

Runs

Data Ingestion

Data Analysis Prac

Retail_sales data for a data agent

Retail_Sales_Data_Agent_KAGISO_MATENCHI...

JOINS

Run selected (1000)

Just now (27s)

workspace: default

New SQL editor: ON

Last edit was 2 m...

Edit

Serverless Starte...

2XS

Schedule

Share

Save*

```

125 INNER JOIN viewing_sessions
126 ON users.user_id=viewing_sessions.user_id;
127
128 --Question 5 : Show users along their subscriptions ,plan names ,and then price ..
129 SELECT user_name,
130        country,
131        plan_name,
132        monthly_price,
133        start_date
134 FROM users
135 INNER JOIN subscriptions ON users.user_id=subscriptions.user_id
136 INNER JOIN plans ON subscriptions.plan_id=plans.plan_id;
137

```

Add parameter

Table

	user_name	country	plan_name	monthly_price	start_date
1	Nomvula	Johannesburg	Basic	75.00	2026-01-15
2	Nomvula	Johannesburg	Premium	199.00	2026-03-10
3	David	Cape Town	Standard	129.00	2026-02-01
4	Anele	Durban	Family	249.00	2026-04-05

4 rows | 27.22s runtime See performance (1) Optimize

Refreshed now

PART B : LEFT JOIN

Question 6:

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Data Analysis Prac

Retail_sales data for a data agent

Retail_Sales_Data_Agent_KAGISO_MATENCHI...

INNER JOINS

LEFT JOIN

Run selected (1000)

Just now (2s)

workspace: default

New SQL editor: ON

Last edit was now

Edit

Serverless Starte...

2XS

Schedule

Share

Save*

```

1 -- Question 6 : Show every user and any subscriptions they have
2 SELECT users.user_id,
3        user_name,
4        subscription_id,
5        start_date
6 FROM users
7 LEFT JOIN subscriptions ON users.user_id = subscriptions.user_id;

```

Add parameter

Table

	user_id	user_name	subscription_id	start_date
1	5	Lerato	null	null
2	1	Nomvula	501	2026-01-15
3	1	Nomvula	503	2026-03-10
4	4	Kabelo	null	null
5	2	David	502	2026-02-01
6	3	Anele	505	2026-04-05

6 rows | 1.80s runtime See performance (1) Optimize

Refreshed now

Question 7:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

+ New Data Analysis Prac Retail_sales data for a data agent Retail_Sales_Data_Agent_KAGISO_MATENCHI... INNER JOINS LEFT JOIN x +

Run selected (1000) Just now (2s) workspace default New SQL editor: ON Last edit was now Serverless Starte... 2XS Schedule Share Save*

```

3      user_name,
4      subscription_id,
5      start_date
6  FROM users
7  LEFT JOIN subscriptions ON users.user_id = subscriptions.user_id;
8
9  -- Question 7 : Show every plan and the subscriptions on it
10 SELECT plans.plan_id,
11        plan_name,
12        subscription_id,
13        user_id
14  FROM plans
15  LEFT JOIN subscriptions ON plans.plan_id = subscriptions.plan_id;

```

Add parameter

	plan_id	plan_name	subscription_id	user_id
1	11	Standard	502	2
2	11	Standard	504	6
3	12	Premium	503	1
4	14	Mobile	null	null
5	13	Family	505	3
6	10	Basic	501	1

6 rows | 1.54s runtime See performance (1) Optimize Refreshed now

Question 8:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

+ New Data Analysis Prac Retail_sales data for a data agent Retail_Sales_Data_Agent_KAGISO_MATENCHI... INNER JOINS LEFT JOIN x +

Run selected (1000) Just now (2s) workspace default New SQL editor: ON Last edit was now Serverless Starte... 2XS Schedule Share Save*

```

12      subscription_id,
13      user_id
14  FROM plans
15  LEFT JOIN subscriptions ON plans.plan_id = subscriptions.plan_id;
16
17 -- Question 8 : Show every show and any viewing sessions on it
18 SELECT shows.show_id,
19        show_title,
20        session_id,
21        watch_minutes
22  FROM shows
23  LEFT JOIN viewing_sessions ON shows.show_id = viewing_sessions.show_id;

```

Add parameter

	show_id	show_title	session_id	watch_minutes
1	703	Tech Tales	902	30
2	706	Wild Earth	null	null
3	704	Cooking Lab	null	null
4	701	Comedy Hour	904	20
5	701	Comedy Hour	901	45
6	702	Crime Time	903	60

6 rows | 1.78s runtime See performance (1) Optimize Refreshed now

Question 9:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

Run selected (1000) Just now (10) workspace default New SQL editor: ON Last edit was now Serverless Start... 2XS Schedule Share Save*

```

21     session_id,
22     watch_minutes
23 FROM shows
24 LEFT JOIN viewing_sessions ON shows.show_id = viewing_sessions.show_id;
25
26 -- Question 9 : Show every viewing session and the user who watched it
27 SELECT session_id,
28        show_id,
29        watch_minutes,
30        viewing_sessions.user_id,
31        user_name
32 FROM viewing_sessions
33 LEFT JOIN users ON viewing_sessions.user_id = users.user_id;

```

Add parameter

Table +

	session_id	show_id	watch_minutes	user_id	user_name
1	902	703	30	2	David
2	901	701	45	1	Nomvula
3	904	701	20	7	null
4	903	702	60	1	Nomvula
5	905	705	90	3	Anele

5 rows | 1.36s runtime See performance (1) Optimize Refreshed now

Question 10:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

Run selected (1000) Just now (20) workspace default New SQL editor: ON Last edit was now Serverless Start... 2XS Schedule Share Save*

```

34
35 -- Question 10 : Show every user, their plan (if any), and the monthly price
36 SELECT user_name,
37        country,
38        plan_name,
39        monthly_price
40 FROM users
41 LEFT JOIN subscriptions ON users.user_id = subscriptions.user_id
42 LEFT JOIN plans ON subscriptions.plan_id = plans.plan_id;
43
44
45
46

```

Add parameter

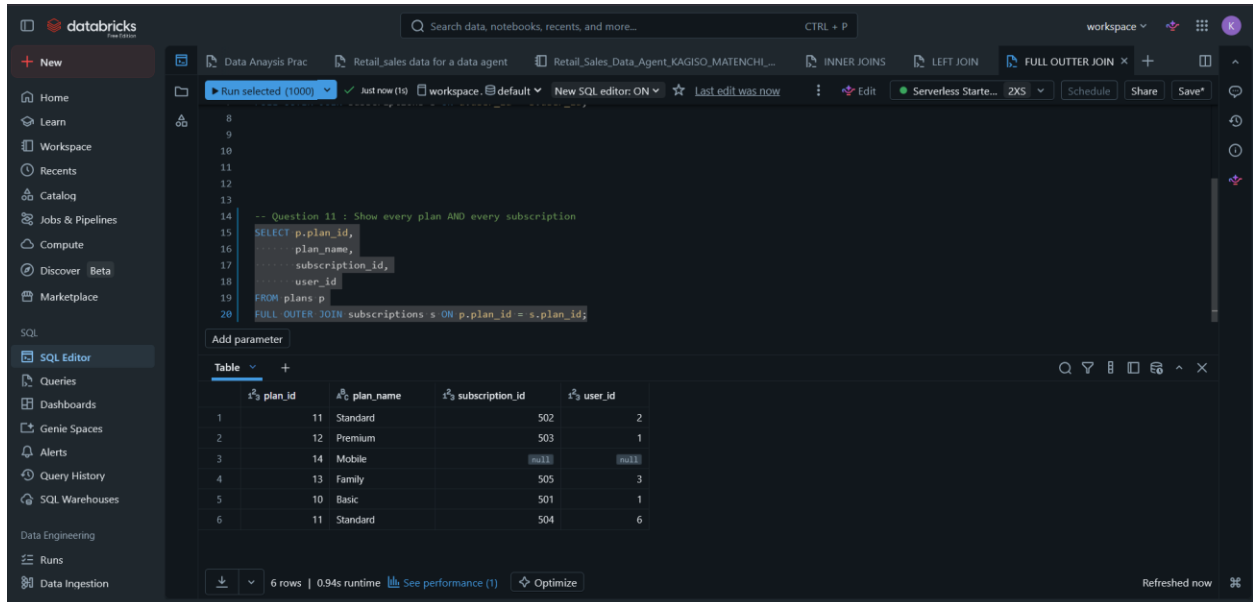
Table +

	user_name	country	plan_name	monthly_price
1	Lerato	Port Elizabeth	null	null
2	Nomvula	Johannesburg	Basic	75.00
3	Nomvula	Johannesburg	Premium	199.00
4	Kabelo	Pretoria	null	null
5	David	Cape Town	Standard	129.00
6	Anele	Durban	Family	249.00

6 rows | 1.82s runtime See performance (1) Optimize Refreshed now

PART C : FULL OUTER JOIN

Question 11:



The screenshot shows the Databricks SQL Editor interface. The SQL query being executed is:

```
-- Question 11 : Show every plan AND every subscription
SELECT p.plan_id,
       plan_name,
       subscription_id,
       user_id
FROM plans p
FULL OUTER JOIN subscriptions s ON p.plan_id = s.plan_id
```

The query results are displayed in a table with 6 rows and 5 columns: `i3 plan_id`, `i3 plan_name`, `i3 subscription_id`, and `i3 user_id`. The results are as follows:

	<code>i3 plan_id</code>	<code>i3 plan_name</code>	<code>i3 subscription_id</code>	<code>i3 user_id</code>
1	11	Standard	502	2
2	12	Premium	503	1
3	14	Mobile	null	null
4	13	Family	505	3
5	10	Basic	501	1
6	11	Standard	504	6

At the bottom of the results, it indicates "6 rows | 0.94s runtime" and provides links for "See performance (1)" and "Optimize". The status "Refreshed now" is also visible.

Question 12:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

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Run selected (1000) Just now (1s) workspace default New SQL editor: ON Last edit was now Serverless Starte... 2XS Schedule Share Save*

```

17     subscription_id,
18     user_id
19 FROM plans p
20 FULL OUTER JOIN subscriptions s ON p.plan_id = s.plan_id;
21
22
23 -- Question 12: Show every plan AND every subscription
24 SELECT p.plan_id,
25        plan_name,
26        subscription_id,
27        user_id
28 FROM plans p
29 FULL OUTER JOIN subscriptions s ON p.plan_id = s.plan_id;

```

Add parameter

	plan_id	plan_name	subscription_id	user_id
1	11	Standard	502	2
2	12	Premium	503	1
3	14	Mobile	null	null
4	13	Family	505	3
5	10	Basic	501	1
6	11	Standard	504	6

6 rows | 0.26s runtime See performance (1) Optimize Refreshed now

Question 13:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

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Run selected (1000) Just now (1s) workspace default New SQL editor: ON Last edit was now Serverless Starte... 2XS Schedule Share Save*

```

26     subscription_id,
27     user_id
28 FROM plans p
29 FULL OUTER JOIN subscriptions s ON p.plan_id = s.plan_id;
30
31
32 -- Question 13 : Show every show AND every viewing session
33 SELECT sh.show_id,
34        show_title,
35        session_id,
36        watch_minutes
37 FROM shows sh
38 FULL OUTER JOIN viewing_sessions vs ON sh.show_id = vs.show_id;

```

Add parameter

	show_id	show_title	session_id	watch_minutes
1	703	Tech Tales	902	30
2	706	Wild Earth	null	null
3	704	Cooking Lab	null	null
4	701	Comedy Hour	904	20
5	702	Crime Time	903	60
6	701	Comedy Hour	901	45
7	null	null	905	90

7 rows | 0.93s runtime See performance (1) Optimize Refreshed now

Question 14 :

Question 14: Show every user AND every viewing session

```

37 FROM shows sh
38 FULL OUTER JOIN viewing_sessions vs ON sh.show_id = vs.show_id;
39
40
41
42 --Question 14 : Show every user AND every viewing session
43 SELECT u.user_id,
44        user_name,
45        session_id,
46        show_id,
47        watch_minutes
48 FROM users u
49 FULL OUTER JOIN viewing_sessions vs ON u.user_id = vs.user_id;

```

user_id	user_name	session_id	show_id	watch_minutes
1	David	902	703	30
2	1 Nomvula	901	701	45
3	3 Anele	904	701	20
4	1 Nomvula	903	702	60
5	3 Anele	905	705	90
6	4 Kabelo	null	null	null
7	5 Lerato	null	null	null

7 rows | 0.99s runtime | See performance (1) | Optimize

Question 15:

Question 15: Show every user, every subscription, and every plan in one query

```

48 FROM users u
49 FULL OUTER JOIN viewing_sessions vs ON u.user_id = vs.user_id;
50
51
52 -- Question 15 : Show every user, every subscription, and every plan in one query
53 SELECT u.user_id,
54        user_name,
55        subscription_id,
56        p.plan_id,
57        plan_name
58 FROM users u
59 FULL OUTER JOIN subscriptions s ON u.user_id = s.user_id
60 FULL OUTER JOIN plans p ON s.plan_id = p.plan_id;

```

user_id	user_name	subscription_id	plan_id	plan_name
1	1 Nomvula	503	12	Premium
2	1 Nomvula	501	10	Basic
3	3 Anele	505	13	Family
4	5 Lerato	null	null	null
5	4 Kabelo	null	null	null
6	2 David	502	11	Standard
7	null	null	504	11 Standard
8	null	null	14	Mobile

8 rows | 1.00s runtime | See performance (1) | Optimize

BONUS CHALLENGE

B1:

databricks Search data, notebooks, recent, and more... CTRL + P workspace

New Data Analysis Prac Retail_sales data for a data agent Retail_Sales_Data_Agent_KAGISO_MATENCHI... INNER JOINS LEFT JOIN FULL OUTER JOIN +

Run selected (1000) Just now (10) workspace: default New SQL editor: ON Last edit was now Edit Serverless Starte... 2XS Schedule Share Save*

```
58 FROM users u
59 FULL OUTER JOIN subscriptions s ON u.user_id = s.user_id
60 FULL OUTER JOIN plans p ON s.plan_id = p.plan_id;
61
62
63 --BONUS CHALLENGE --
64
65 -- Bonus 1 : Which users have NOT subscribed to any plan?
66 SELECT u.user_id,
67        user_name
68 FROM users u
69 LEFT JOIN subscriptions s ON u.user_id = s.user_id
70 WHERE s.subscription_id IS NULL;
```

Add parameter

	user_id	user_name
1	5	Lerato
2	4	Kabelo

2 rows | 0.07s runtime See performance (1) Optimize Refreshed now

B2:

databricks Search data, notebooks, recent, and more... CTRL + P workspace

New Data Analysis Prac Retail_sales data for a data agent Retail_Sales_Data_Agent_KAGISO_MATENCHI... INNER JOINS LEFT JOIN FULL OUTER JOIN +

Run selected (1000) Just now (10) workspace: default New SQL editor: ON Last edit was now Edit Serverless Starte... 2XS Schedule Share Save*

```
66 SELECT u.user_id,
67        user_name
68 FROM users u
69 LEFT JOIN subscriptions s ON u.user_id = s.user_id
70 WHERE s.subscription_id IS NULL;
71
72 --Bonus 2 : Which subscriptions reference users that do NOT exist in the users table?
73 SELECT s.subscription_id,
74        s.user_id,
75        start_date
76 FROM subscriptions s
77 LEFT JOIN users u ON s.user_id = u.user_id
78 WHERE u.user_id IS NULL;
```

Add parameter

	subscription_id	user_id	start_date
1	504	6	2026-03-20

1 row | 0.74s runtime See performance (1) Optimize Refreshed now

B3:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

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Run selected (1000) Just now (1) workspace default New SQL editor: ON Last edit was now Edit Serverless Starte... 2XS Schedule Share Save*

```

74      s.user_id,
75      start_date
76  FROM subscriptions s
77  LEFT JOIN users u ON s.user_id = u.user_id
78  WHERE u.user_id IS NULL;
79
80  -- Bonus 3 : Which shows have NEVER been watched?
81  SELECT sh.show_id,
82         show_title,
83         genre
84  FROM shows sh
85  LEFT JOIN viewing_sessions vs ON sh.show_id = vs.show_id
86  WHERE vs.session_id IS NULL;

```

Add parameter

	show_id	show_title	genre
1	706	Wild Earth	Documentary
2	704	Cooking Lab	Lifestyle

2 rows | 1.29s runtime See performance (1) Optimize Refreshed now

B4:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

New Home Learn Workspace Recents Catalog Jobs & Pipelines Compute Discover *Beta* Marketplace SQL SQL Editor Queries Dashboards Genie Spaces Alerts Query History SQL Warehouses Data Engineering Runs Data Ingestion

Run selected (1000) Just now (<1s) workspace default New SQL editor: ON Last edit was now Edit Serverless Starte... 2XS Schedule Share Save*

```

82      show_title,
83      genre
84  FROM shows sh
85  LEFT JOIN viewing_sessions vs ON sh.show_id = vs.show_id
86  WHERE vs.session_id IS NULL;
87
88  -- Bonus 4 : Which viewing sessions reference shows that do NOT exist?
89  SELECT vs.session_id,
90         vs.show_id,
91         watch_minutes
92  FROM viewing_sessions vs
93  LEFT JOIN shows sh ON vs.show_id = sh.show_id
94  WHERE sh.show_id IS NULL;

```

Add parameter

	session_id	show_id	watch_minutes
1	905	705	90

1 row | 0.33s runtime See performance (1) Optimize Refreshed now

B5:

The screenshot displays the Databricks SQL Editor interface. The top navigation bar includes the Databricks logo, a search bar, and a workspace dropdown. The left sidebar contains a navigation menu with options like Home, Learn, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover (Beta), Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Spaces, Alerts, Query History, SQL Warehouses, Data Engineering, Runs, and Data Ingestion. The main editor area shows a SQL query with line numbers 90 to 102. The query is as follows:

```
90 vs.show_id,  
91 watch_minutes  
92 FROM viewing_sessions vs  
93 LEFT JOIN shows sh ON vs.show_id = sh.show_id  
94 WHERE sh.show_id IS NULL;  
95  
96 -- Bonus 5 : Which plans have NO subscribers?  
97 SELECT p.plan_id,  
98        plan_name,  
99        monthly_price  
100 FROM plans p  
101 LEFT JOIN subscriptions s ON p.plan_id = s.plan_id  
102 WHERE s.subscription_id IS NULL;
```

Below the query editor, there is a table view showing the results of the query. The table has three columns: plan_id, plan_name, and monthly_price. The results are as follows:

plan_id	plan_name	monthly_price
14	Mobile	55.00

The bottom status bar indicates that the query ran for 1.11s and shows an 'Optimize' button.

SQL Environment: All SQL queries in this exercise were written and executed in **Databricks**, using the SQL editor with a Serverless Starter compute instance. Databricks uses **Apache Spark SQL**, which supports standard SQL syntax including INNER JOIN, LEFT JOIN, and FULL OUTER JOIN.